Revisiting Lifetime-Aware VM Allocation

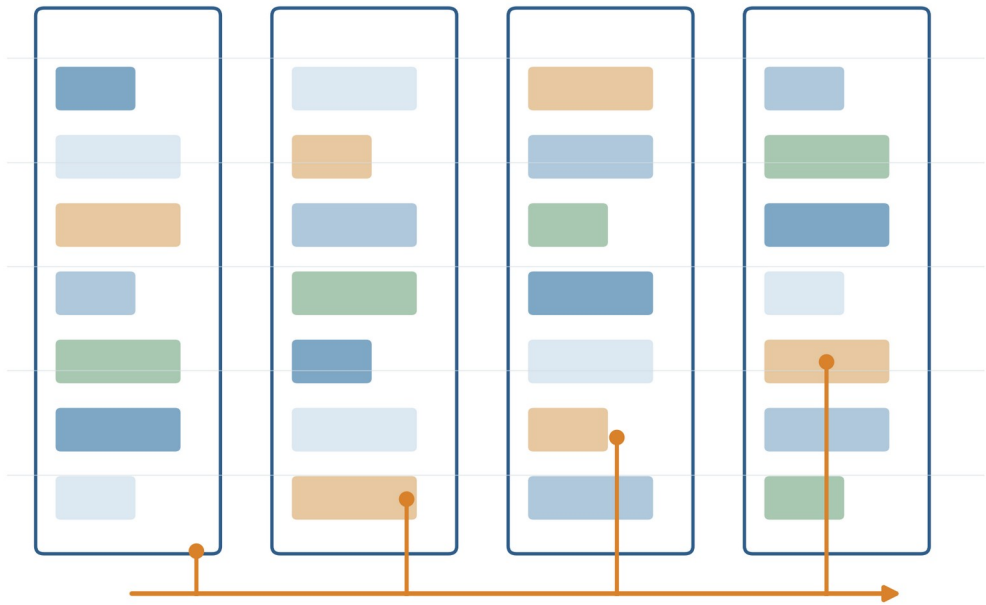
Reconstruction, Diagnosis and Prior-Fused Lifetime Prediction

Lifetime prediction, confidence routing and prior fusion

Synthetic reconstruction

100 fresh test seeds

Pre-registered, Holm-corrected



CE7490 Course Project - 26S1 Group 5

Zhang Qi (G2510784F) qi015@e.ntu.edu.sg
Zhang Saining (G2511061H) saining002@e.ntu.edu.sg
Zheng Yitao (G2608664B) YZHENG056@e.ntu.edu.sg

02

Three questions, three answers

RQ1 · Prediction quality

Does LAVA's benefit depend on how accurate lifetime predictions are?

−0.19 → +0.51 pp

Yes. LAVA loses to Best-Fit at 25% accuracy and gains clearly at 100%.

RQ2 · Confidence routing

Does routing low-confidence VMs away from LAVA help?

−0.090 pp

No. Host-level routing is below Best-Fit; three measured causes contribute.

RQ3 · Prior fusion

Can the lifetime prediction itself be made robust?

3 of 4

Yes. Beats LAVA in 3 of 4 pre-registered tests and no longer loses at low accuracy.

Lesson: uncertainty helped inside the lifetime estimate, not as a policy switch.

Fresh test seeds 1000–1099 (n = 100); all values relative to same-seed Best-Fit.

03

VM placement and resource fragmentation



Multidimensional fragmentation

CPU and memory can each be free but unusable together.

Time is the hidden dimension

VMs that fit equally well can exit at very different times.

Operational consequence

Empty hosts support maintenance, larger VMs and energy management.

Objective: maximize the time-weighted share of empty hosts without rejecting any VM.

Scale: about 90% of host-time is empty under every policy (low load, long drain tail); 1 pp ≈ 1.8 of 180 hosts.

04 LAVA prediction and host lifecycle

Distributions

Estimate a remaining-lifetime distribution for each VM.

Re-prediction

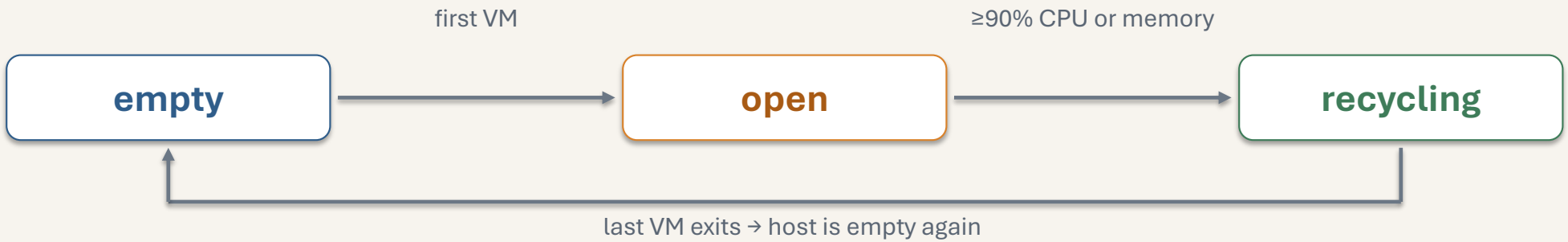
Condition on uptime to revise the expected remaining lifetime.

Host lifecycle

Empty, open and recycling states guide placement.

Temporal placement

Align departures and fill recycling hosts with shorter VMs.



Adaptation to mistakes

Residual cohorts lower the host class when they leave; missed deadlines raise it when VMs outlive their prediction.

Ling et al., “LAVA: Lifetime-Aware VM Allocation with Learned Distributions and Adaptation to Mispredictions”, MLSys 2025 (arXiv:2412.09840).

05

Reconstruction scope and seed splits

Recovered settings

180 hosts, 2,500 VMs per trace, 8 arrivals per hour, seed splits and metric.

Inferred defaults

Mixture parameters, VM shapes, scoring details and the controlled confidence model.

Unrecoverable inputs

Original traces, random streams, code history and per-seed results.

Split	Seeds	Used for
Initial sweep	0–19	RQ1 accuracy sweep; development of the new method
Validation	200–205	Report's threshold selection; τ = 0.3 kept frozen
Held-out	20–39	Report's test split; rerun with all ten policies
Fresh test	1000–1099	Pre-registered evaluation of prior fusion (n = 100)
Robustness	2000–2049	Predictor variants, same-seed control and load × 2 (n = 50)
Threshold sensitivity	200–219	Descriptive τ scan for the new method (n = 20)

06

Methods compared

Best-Fit

Resource-only baseline: minimize residual CPU and memory.

LA-Binary

One-shot short/long label at arrival, then Best-Fit.

NILAS

Re-predicted temporal extension cost, then Best-Fit.

LAVA

Lifetime classes and host states; adapts when VMs outlive their prediction.

VM-Level CA (VM-CA)

Per-arrival confidence gate between LAVA and Best-Fit on shared hosts.

Host-Level CA (Host-CA)

Same gate, but each host keeps one mode until it empties.

New in this project: *-Bayes variants (RQ3)

NILAS-Bayes, LAVA-Bayes, VM-CA-Bayes and Host-CA-Bayes run exactly the same scheduling code as their base policy and only swap in the prior-fused lifetime predictor.

07

Prediction settings and information boundary

Correct bucket

Log10 error with σ = 0.001.

Incorrect bucket

Log10 error with σ = 3 (orders of magnitude).

Information boundary

The scheduler never sees the bucket or the true lifetime.

Controlled accuracy = share of VMs in the correct bucket

25% — 50% — 75% — 85% — 100%

Default predictor

Profile-level two-exponential mixtures give the conditional remaining lifetime and a class probability. Numeric parameters are inferred.

Oracle reference

True remaining lifetime; a full-information reference, not an upper bound.

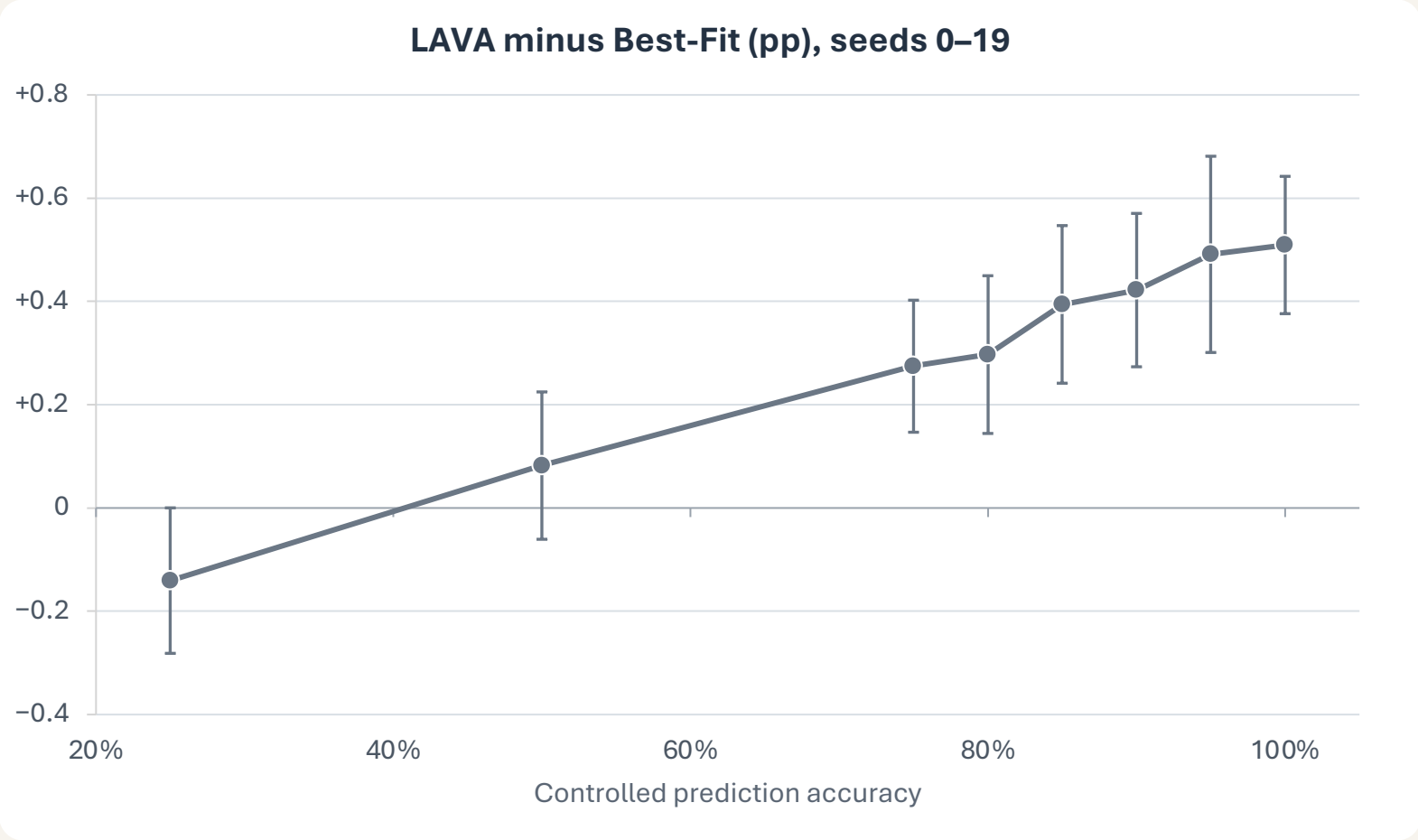
Statistical protocol

Identical traces and noise for all policies; paired Student-t 95% intervals.

***-Bayes adds only the profile prior, which the default predictor already uses.**

08

RQ1: Prediction accuracy drives LAVA's gain



First significant point

75% accuracy: +0.275 [+0.147, +0.403] pp

Default predictor

+0.124 pp here; not replicated on held-out (+0.001) or fresh seeds (−0.043).

Oracle reference

+0.575 [+0.433, +0.718] pp

Initial split (n = 20 per point), paired 95% intervals. The 75% crossing point is specific to this reconstruction.

09

Held-out sensitivity and the RQ2 hypothesis

+0.001 pp

LAVA, default predictor (held-out 20–39)

+0.661 pp

LAVA with true lifetimes (held-out 20–39)

−0.187 pp

LAVA at 25% accuracy (fresh seeds)

Observation

LAVA gains little with the default predictor and loses to Best-Fit when predictions are poor.

→

Hypothesis

The earlier report's idea: send low-confidence requests to Best-Fit, per VM or per host.

→

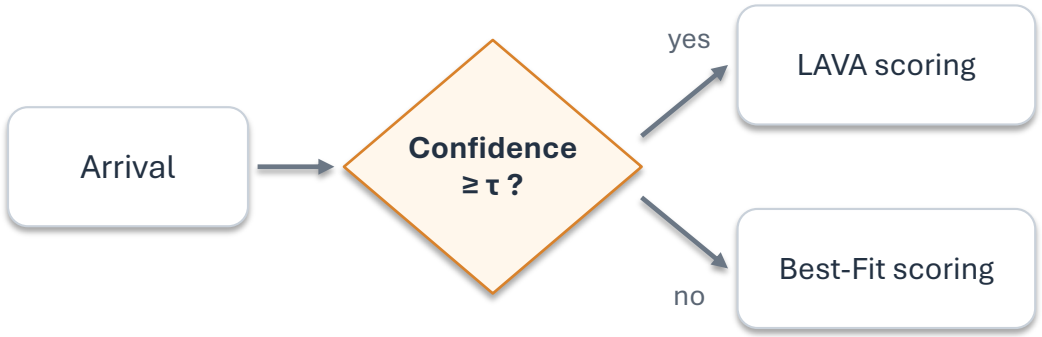
Constraint

Workload and threshold stay fixed; no search for a result that matches the earlier report.

All values are paired differences relative to same-seed Best-Fit.

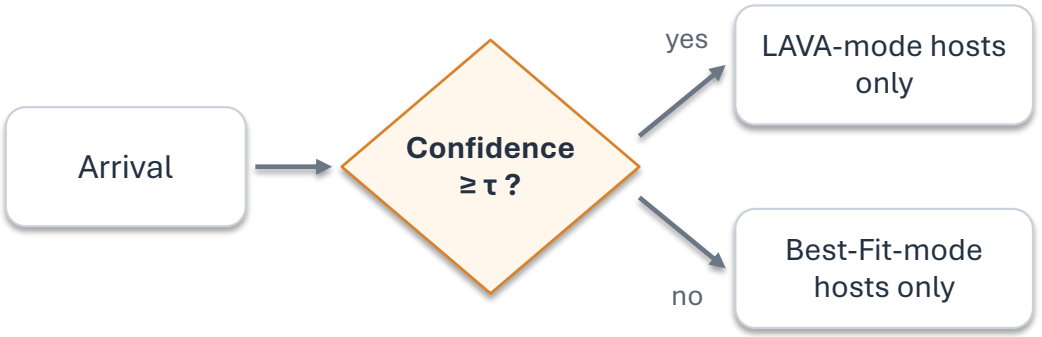
RQ2: Two ways to route by confidence

VM-Level CA



Both decisions land on one shared set of hosts: modes mix on a host.

Host-Level CA



A host keeps its first mode until empty; cross-mode only if nothing else fits.

608 + 319 → 0

cross-mode placements per trace, VM-CA → Host-CA

20%

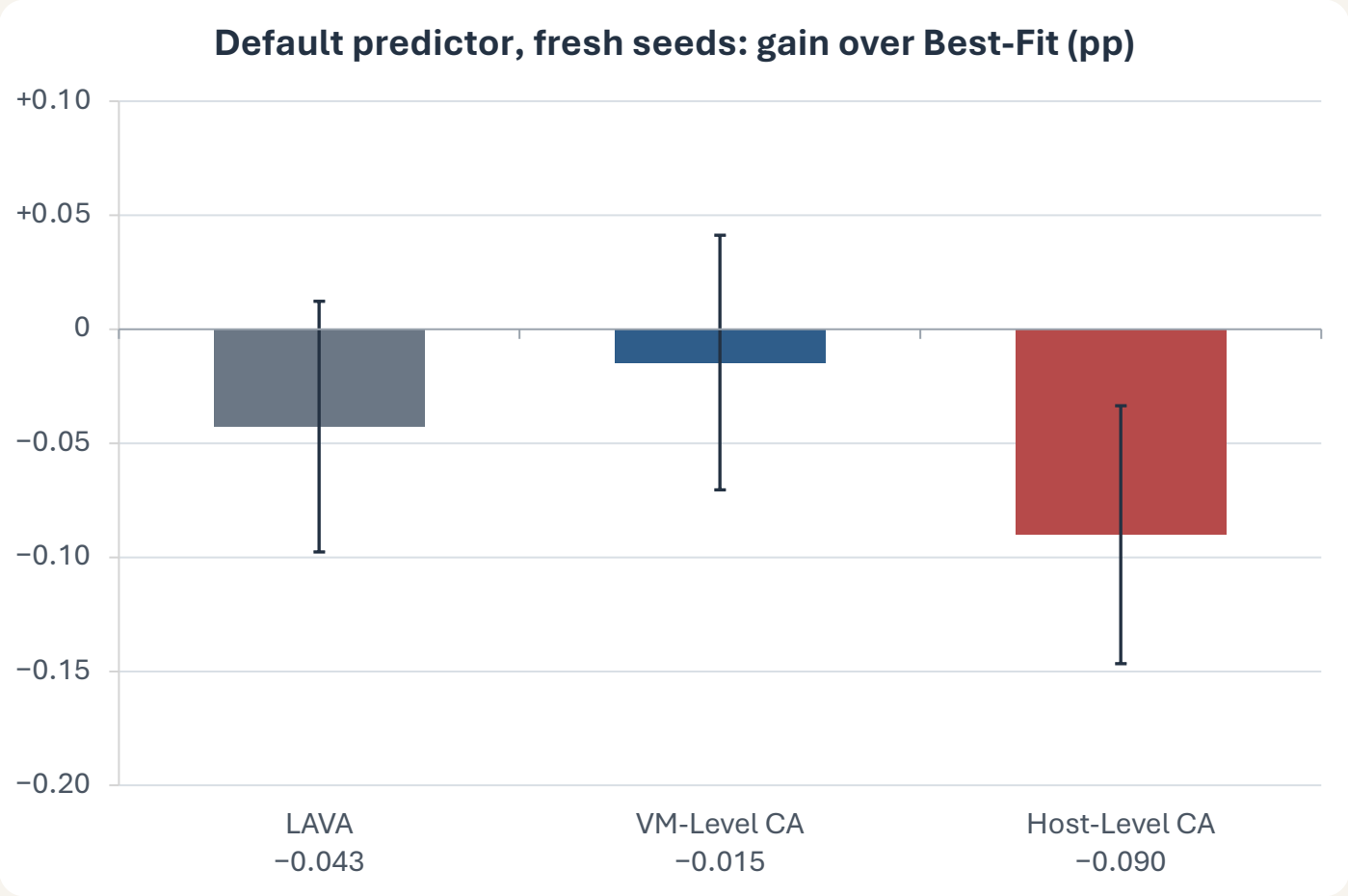
of arrivals pass the gate, identical in both variants

0

capacity fallbacks in any run

Mechanism check on held-out seeds 20–39, default predictor: isolation does exactly what it was designed to do.

Diagnostic negative result: routing does not beat Best-Fit



Fresh seeds (n = 100)

Host-CA − Best-Fit: −0.090 [−0.147, −0.034] pp, significantly below zero.

Held-out seeds 20–39

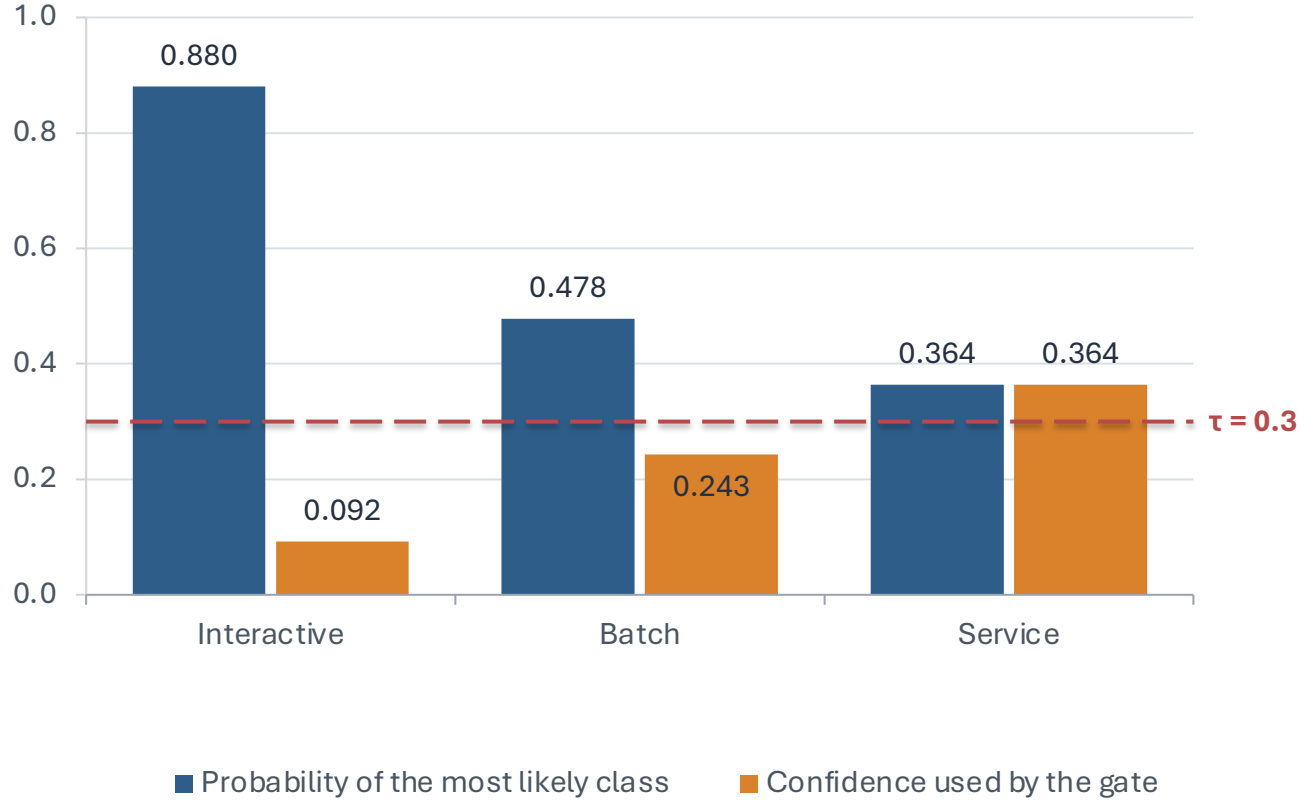
Host-CA −0.045 vs Best-Fit, −0.158 vs VM-CA; VM-CA's +0.114 did not replicate.

Surviving report (historical)

Reported Host-CA − Best-Fit: +0.241 [+0.125, +0.357] pp; not reproduced here.

Paired 95% intervals; threshold τ = 0.3 as in the report.

Why confidence routing does not help here



Inverted confidence

Interactive VMs are 88% likely to exit within 1 h but get 0.092; service VMs get 0.364.

Blind gate

Under controlled noise ≥99.96% of arrivals pass τ = 0.3, so both variants essentially reduce to LAVA.

Isolation costs packing

Host-CA loses 0.99 pp of empty hosts during arrivals; the tail recovers about 72%.

Default predictor at arrival; gate rates and windows on fresh seeds. Fixing the inversion alone did not help on development seeds either.

RQ3: Prior-fused lifetime prediction

Inputs

Profile prior, noisy value, accuracy and σ

Posterior over T

Log-grid atoms; values at the 336 h cap are censored

Uptime filter

Keep only atoms above the current age

Outputs

Mean remaining lifetime and class confidence

Scheduler

NILAS, LAVA and CA rules unchanged

Implausible values fall back

Interactive VM at 25% accuracy predicted 300 h: the original predictor uses 300 h, the fused one 1.52 h (prior mean 1.07 h).

Censoring at the cap

A value at 336 h means “at least 336 h”. At uptime 300 h the original expects 36 h left, the fused one 151 h.

No overdue jump

A VM that outlives its prediction keeps a posterior estimate instead of jumping to “remaining = uptime”.

Same information as the original confidence model plus the profile prior; never the bucket or the true lifetime.

Implementation: BayesPredictor in lava/model.py (44 lines), unit-tested in tests/test_bayes.py; examples computed by the code.

Pre-registered evaluation protocol

1 Develop

Seeds 0–19 only. Two other ideas rejected there.

2 Register

28 Oct, 20:10 (UTC+8): hypotheses, seeds, τ, decision rule and code hashes frozen.

3 Test

Fresh seeds 1000–1099 (n = 100); τ = 0.3 from the report, nothing tuned.

4 Infer

Eight paired t-tests, Holm–Bonferroni at α = 0.05.

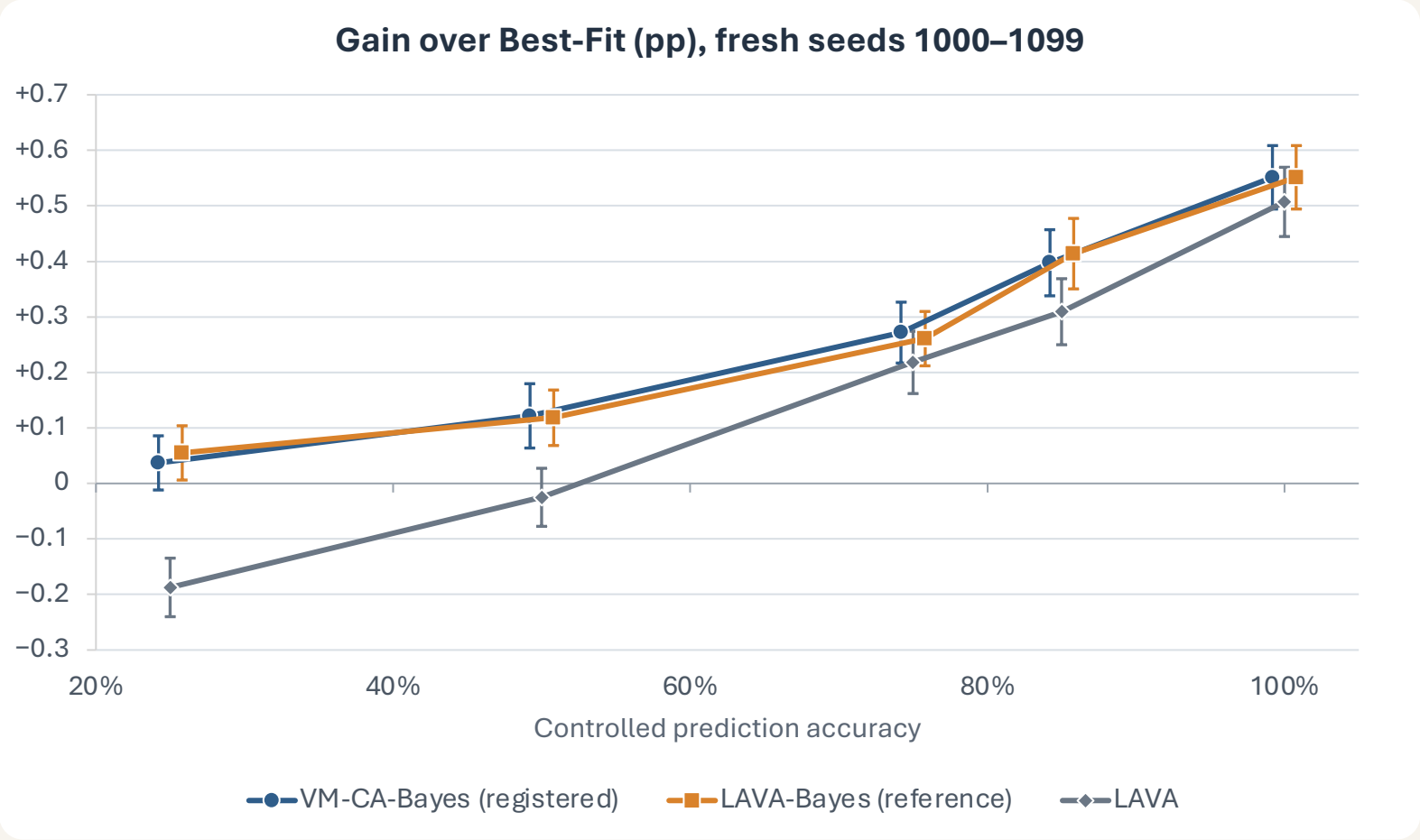
Compute

23,240	64	≈6 min	0
simulations	low-priority CPU processes on the H20 server	wall time; no GPU needed	scheduling failures

Also run: robustness seeds 2000–2049 (control and load × 2 added after review), τ scan on 200–219, and the report's held-out seeds 20–39.

Only the eight registered tests are multiplicity-adjusted; everything else is descriptive.

Fusion removes LAVA's low-accuracy penalty



Registered: vs LAVA

+0.224, +0.147, +0.054, +0.088 pp at 25–85%; 3 of 4 hold.

Registered: vs Best-Fit

Holds at 50–85%; at 25%: +0.037 pp, not significant.

Oracle reference

LAVA with true lifetimes: +0.552 pp

Whiskers: unadjusted 95% intervals (n = 100). LAVA-Bayes was not registered and is shown for reference.

Pre-registered results: 3 of 4 hold against LAVA

Accuracy	VM-CA-Bayes – LAVA	Holm p	Holds	VM-CA-Bayes – Best-Fit	Holm p	Holds
25%	+0.224 [+0.174, +0.275]	<0.001	yes	+0.037 [−0.012, +0.086]	0.14	no
50%	+0.147 [+0.088, +0.205]	<0.001	yes	+0.122 [+0.064, +0.180]	<0.001	yes
75%	+0.054 [−0.001, +0.109]	0.11	no	+0.272 [+0.217, +0.327]	<0.001	yes
85%	+0.088 [+0.037, +0.139]	0.003	yes	+0.398 [+0.338, +0.457]	<0.001	yes

Not supported vs LAVA at 75%

+0.054 [−0.001, +0.109] pp: the interval includes zero (Holm p = 0.11).

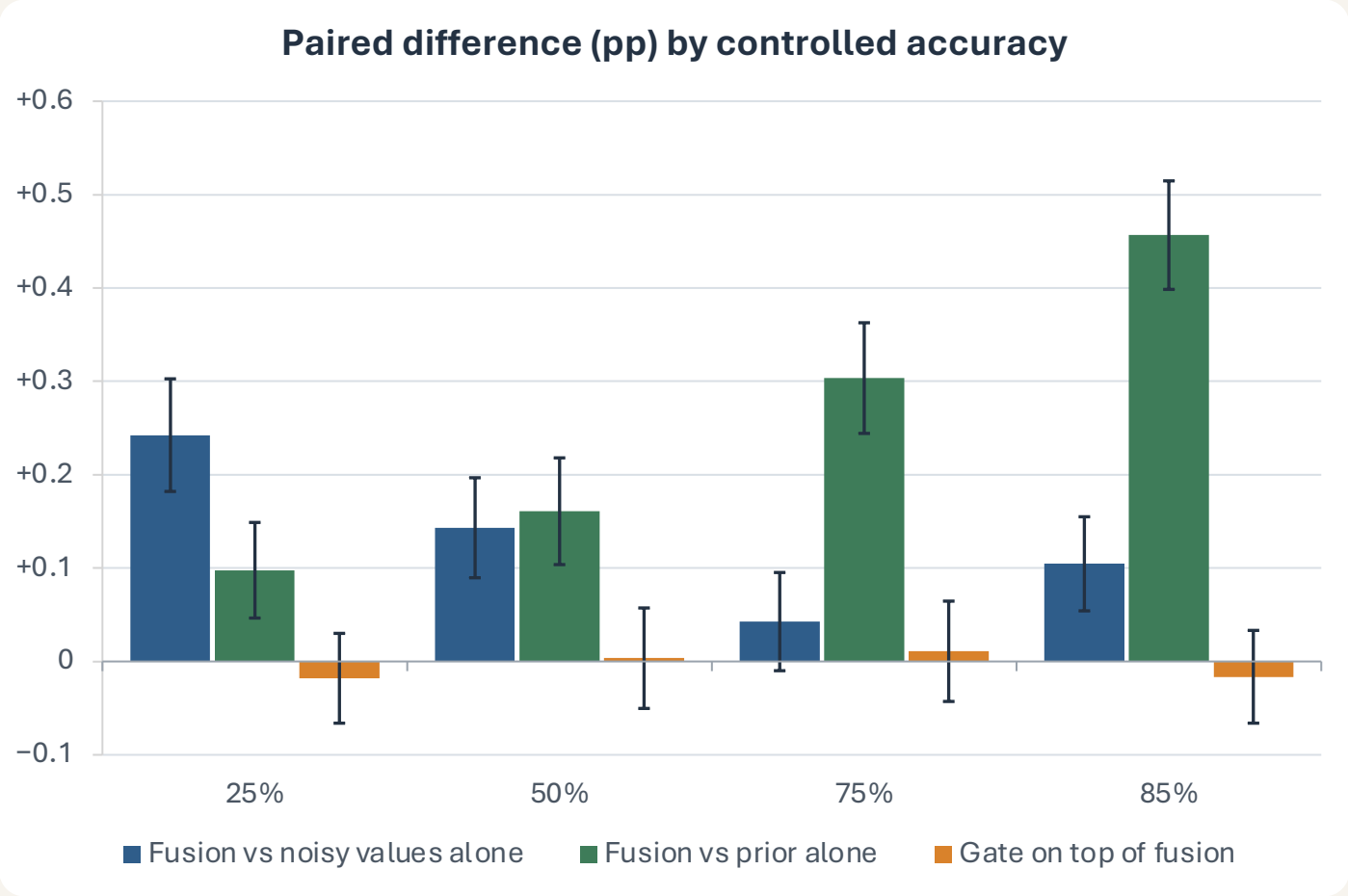
Not supported vs Best-Fit at 25%

+0.037 [−0.012, +0.086] pp. Secondary, unadjusted: NILAS-Bayes +0.084 ($p < 0.001$), LAVA-Bayes +0.055 ($p = 0.03$).

Decision rule fixed before testing: mean > 0 and Holm-adjusted $p < 0.05$. Intervals are unadjusted 95% CIs.

Plain LAVA would also pass the Best-Fit tests at 75% and 85%, so those two say little about fusion itself.

Post hoc: the gain comes from the estimate, not the gate



Fusion vs its ingredients

+0.10 to +0.46 pp over the prior alone; +0.04 to +0.24 over noisy values alone (n.s. at 75%).

Gate on top

−0.02 to +0.01 pp; every 95% interval includes zero (half-width ≈ 0.05). The development-seed gain did not replicate.

Weakly selective gate

Routes 82–87% of correct and 73–75% of wrong predictions to LAVA.

Post hoc, not registered; fresh seeds (n = 100). “Prior alone” = the same seeds under the default predictor.

Robustness and the failure mode

VM-CA-Bayes – LAVA (pp)	25%	50%	75%	85%
Protocol, same seeds	+0.225	+0.185	+0.113	+0.127
Incorrect bucket σ = 1	+0.024	+0.102	+0.025	+0.093
Incorrect bucket σ = 0.5	+0.070	+0.062	+0.037	+0.092
Fusion assumes σ = 3 (true 1)	+0.083	+0.074	+0.002	+0.066
Fusion assumes σ = 1 (true 3)	+0.168	+0.173	+0.109	+0.069
Prior means × 0.5	+0.267	+0.135	−0.033	−0.116
Prior means × 2	+0.223	+0.215	+0.149	+0.112
Arrival rate × 2	+0.476	+0.271	+0.177	+0.101

Failure mode: short prior

Prior means × 0.5 at 85% is the only significantly negative cell (−0.116 pp; 1 of 32).

Load × 2 (16 VMs/h)

Gain over LAVA rises to +0.476, +0.271, +0.177 pp at 25%, 50% and 75%.

Threshold τ

0.1–0.5: positive mean at every accuracy; τ ≥ 0.7 loses the gain at 50–85% (backup slide).

Seeds 2000–2049 (n = 50), descriptive. Green / red: 95% interval above / below zero.

Same-seed control and load × 2 were added after review; the registered 90-host variant only rescales the 180-host run and is omitted.

Findings and limits

What we found

RQ1: prediction quality is first-order

LAVA moves from −0.187 to +0.507 pp between 25% and 100% accuracy.

RQ2: a diagnostic negative result

Inverted confidence, a blind gate and the cost of isolation each contribute to the missing gain.

RQ3: prior fusion works

Beats LAVA in 3 of 4 registered tests; post hoc, the estimate rather than the gate appears to carry the gain.

Limits

Synthetic workload: about 20% offered load; the drain tail is about two thirds of the metric.

Calibrated prior: the generating distribution; halved means fail at 85%.

Reconstruction: NILAS beats LAVA at most accuracies here; original code and traces were lost.

Statistics: registration is self-timestamped; robustness cells are descriptive.

Lesson: uncertainty helped inside the lifetime estimate, not as a policy switch.

Thank you

Questions?

−0.19 → +0.51 pp

RQ1: LAVA vs Best-Fit, 25% → 100% accuracy

−0.090 pp

RQ2: Host-Level CA vs Best-Fit

3 of 4

RQ3: registered wins against LAVA

Report: Revisiting_Lifetime_Aware_VM_Allocation_Report_v3.pdf · Artifact: lava_reproduction/ (code, raw results, registration).

Backup: threshold sensitivity

τ	25%	50%	75%	85%
0.1	+0.208 [+0.079, +0.338]	+0.139 [+0.023, +0.256]	+0.109 [−0.024, +0.241]	+0.139 [+0.021, +0.258]
0.2	+0.268 [+0.185, +0.352]	+0.112 [+0.002, +0.222]	+0.135 [+0.002, +0.268]	+0.109 [+0.023, +0.194]
0.3	+0.274 [+0.156, +0.392]	+0.192 [+0.072, +0.312]	+0.130 [−0.021, +0.281]	+0.157 [+0.063, +0.252]
0.5	+0.172 [+0.093, +0.252]	+0.074 [−0.041, +0.189]	+0.039 [−0.076, +0.154]	+0.142 [+0.002, +0.281]
0.7	+0.197 [+0.086, +0.308]	−0.037 [−0.165, +0.090]	−0.125 [−0.240, −0.011]	−0.090 [−0.196, +0.016]
0.8	+0.124 [+0.040, +0.208]	−0.005 [−0.106, +0.096]	−0.113 [−0.256, +0.029]	−0.148 [−0.234, −0.062]
0.9	+0.180 [+0.074, +0.287]	−0.023 [−0.130, +0.084]	−0.094 [−0.228, +0.040]	−0.104 [−0.198, −0.011]

VM-CA-Bayes minus LAVA (pp), seeds 200–219 (n = 20), descriptive; τ = 0.3 was frozen from the report, not chosen here.

At τ ≥ 0.7 the gate still sends 64–80% of arrivals to LAVA at 75–85% accuracy (81–85% at τ = 0.3).

Backup: all policies versus Best-Fit (fresh seeds)

Setting	LAVA	NILAS	VM-CA	Host-CA	LAVA-B	NILAS-B	VM-CA-B	Host-CA-B
25%	−0.187	−0.055	−0.178	−0.180	+0.055	+0.084	+0.037	−0.271
50%	−0.025	+0.076	−0.025	−0.025	+0.118	+0.135	+0.122	−0.162
75%	+0.218	+0.218	+0.218	+0.218	+0.261	+0.324	+0.272	+0.030
85%	+0.309	+0.349	+0.309	+0.309	+0.414	+0.386	+0.398	+0.199
100%	+0.507	+0.476	+0.507	+0.507	+0.551	+0.599	+0.551	+0.551
Default	−0.043	+0.049	−0.015	−0.090	−0.043	+0.049	−0.015	−0.090
Oracle	+0.552	+0.581	+0.552	+0.552	+0.552	+0.581	+0.552	+0.552

Mean pp versus same-seed Best-Fit (n = 100); “-B” = prior-fused variant. Green / red: 95% interval above / below zero.

Under the default predictor and the Oracle the fused variants equal their base policies by construction.